4.1 Related Rates

Example 4.1.1. *How fast is the area of a rectangle changing if one side is 10cm long and is increasing at a rate of 2cm/s and the other side is 8cm long and is decreasing at a rate of 3cm/s?*

Example 4.1.2. *How fast is the surface area of a ball changing when the volume of the ball is* $32\pi/3$ *cm*^{3} *and is increasing at 2cm*^{3}*/s? (The surface are of the ball is* $A = 4\pi r^2$ *and the volume is* $V = \frac{4}{3}\pi r^3$*. Note that* $V(r) = \int_0^r A(r)dr$*)*

Example 4.1.3. *A point is moving to the right along the first quadrant portion of the curve* $x^2y^3 = 72$. *When the point has coordinates* $(3,2)$, *its horizontal velocity is 2 units/s. What is its vertical velocity?*

Exercises.

1. *Find the rate of change of the area of a square whose side is 6 cm long, if the side length is increasing at 2 cm/min.*

2. *Air is being pumped into a spherical balloon. The volume of the balloon is increasing at a rate of 20 cm^3/s when the radius is 30 cm. How fast is the radius increasing at that time? (The volume of a ball of radius r is* $V = \frac{4}{3}\pi r^3$*.)*

3. *The area of a circle is decreasing at a rate of 2 cm^2/min. How fast is the radius of the circle changing when the area is 100 cm^2?*

4.2 Indeterminate Forms

To evaluate the limit $\lim_{x\to 0}\frac{\sin x}{x}$ we can not plug in $x=0$. We call $\sin x/x$ an **indeterminate form** of $[0/0]$ at $x=0$.

The limit of an indeterminate form $[0/0]$ can be any number.

$$\lim_{x\to 0}\frac{x}{x}=1, \qquad \lim_{x\to 0}\frac{x}{x^3}=\infty, \qquad \lim_{x\to 0}\frac{x^3}{x^2}=0.$$

There are other types of indeterminate forms $[\infty/\infty]$, $[0\cdot\infty]$, $[\infty-\infty]$, $[0^\infty]$, $[\infty^0]$, $[1^\infty]$.

Theorem 4.2.1 (l'Hopital's Rules)**.** *Let f and g are differentiable on an interval containing a. Suppose that $\lim_{x\to a} f(x)$ and $\lim_{x\to a} g(x)$ are either both 0 or both $\pm\infty$. If $\lim_{x\to a}\frac{f'(x)}{g'(x)}$ exists then*

$$\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}.$$

The results hold true if $\lim_{x\to a}$ is replaced by $\lim_{x\to a+}$ and $\lim_{x\to a-}$ or if $a=\pm\infty$.

Example 4.2.1. *Evaluate*

$$\lim_{x\to 1} \frac{\ln x}{x^2 - 1}$$

Example 4.2.2. *Evaluate*

$$\lim_{x \to 0} \frac{2\sin x - \sin(2x)}{2e^x - 2 - 2x - x^2}$$

Example 4.2.3.

$$\lim_{x \to 1+} \frac{x}{\ln x}$$

Example 4.2.4.

$$\lim_{x \to 0+} \frac{1}{x} - \frac{1}{\sin x}$$

Example 4.2.5.

$$\lim_{x \to 0+} x^x.$$

Example 4.2.6. *Evaluate*

$$\lim_{x\to\infty}\left(1+\sin\frac{3}{x}\right)^{x}$$

Exercises.

1. $\lim_{x\to\infty} \frac{x^2}{e^x}$.

   Answer: 0

2. $\lim_{x\to 0} \dfrac{x-\sin x}{x^3}$

   Answer: $1/6$

3. $\lim_{x\to 0} \dfrac{x-\sin x}{x-\tan x}$

   Answer: $-\frac{1}{2}$

4. $\lim_{x\to 0+} x^{\sqrt{x}}$

   Answer: 1

5. $\lim_{x\to 1} \frac{\ln(ex)-1}{\sin \pi x}$

   Answer: $-\frac{1}{\pi}$

6. $\lim_{x\to 0+} \frac{\csc x}{\ln x}$

   Answer: ∞

7. $\lim_{x\to 0} (1+\tan x)^{1/x}$

   Answer: 1

4.3 Extreme Values

A function has an **absolute maximum value** $f(x_0)$ if $f(x) < f(x_0)$ holds for every x in its domain.

Similarly, define **absolute minimum value**.

If it has an absolute min/max, then that value may be achieved at more than one point. For example the function $\cos x$ attains its absolute max at $x = 2n\pi$ for any integer n.

A function may or may not have an absolute min/max value. For example the function $f(x) = x, 0 < x < 1$ does not have an absolute maximum or minimum.

Recall from the section on continuous functions that,

```
A continuous function defined on a closed and bounded interval must have an absolute maximum and an absolute minimum.
```

Maximum and minimum values of a function are collectively referred to as **extreme values**.

Function f has a **local maximum** value $f(x_0)$ if there exists $h > 0$ such that $f(x) \leq f(x_0)$ whenever x is in the domain of f and $|x - x_0| < h$.

Similarly we define **local minimum**.

We define **critical points** of f where $f'(x) = 0$, **singular points** of f where x is in domain of f and $f'(x)$ does not exist.

Following theorem says where the extreme values are located.

Theorem 4.3.1. *If the function f is defined on an interval I and has a local max or local min at $x = x_0$ then x_0* ***must*** *be either a critical point, a singular point or an endpoint of the interval.*

Example 4.3.1. *Find the maximum and minimum values of the function $g(x) = x^3 - 3x^2 - 9x + 2$ on the interval $-2 \le x \le 2$.*

Example 4.3.2. *Find the maximum and minimum values of* $h(x) = 3x^{2/3} - 2x$ *on the inteval* $[-1, 1]$.

The first derivative test

By investigating the sign of the first derivative we can determine whether an extrema is a local minimum or local maximum.

Example 4.3.3. *Find the local and absolute extreme values of* $f(x) = x^4 - 2x^2 - 3$ *on the interval* $[-2, 2]$. *Sketch the graph of* f.

Example 4.3.4. *Locate all extreme values of $f(x) = x\sqrt{2 - x^2}$. Determine whether any of these extreme values are absolute. Sketch the graph.*

4.4 Concavity and Inflections

We say f is **concave up** on an interval I if f' is increasing on I and **concave down** on I if f' decreasing on I.

Note that if f is concave up then f lies above its tangents and below its chords while if f is concave down then f lies below its tangents and above its chords.

If f changes its concavity at x_0 then we call x_0 and **inflection point**.

Theorem 4.4.1. *Assume f is twice differentiable.*

a) If $f'' > 0$ on an interval I then f is concave up on I,

b) If $f'' < 0$ on an interval I then f is concave down on I,

c) If f has an inflection point at x_0 then $f''(x_0) = 0$.

Note $f''(x_0) = 0$ does not necessarily mean x_0 is an inflection point, for example for $f(x) = x^4$ $f''(0) = 0$ while f does not change concavity at $x = 0$.

Example 4.4.1. *Determine the intervals of concavity of* $f(x) = x^6 - 10x^4$ *and the inflection points of its graph.*

Example 4.4.2. *Determine the intervals of increase, decrease , the local extreme values and the concavity of* $f(x) = x^4 - 2x^3 + 1$. *Sketch the graph of* f.

The Second Derivative Test

Theorem 4.4.2. *a) If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a local max at x_0.*

b) If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a local min at x_0.

c) If $f'(x_0) = f''(x_0)$, then no conclusion can be drawn.

Example 4.4.3. *Find an classify the critical points of* $f(x) = x^2 e^{-x}$.

4.5 Graphs of Functions

Definition 4.5.1. *The graph of $y = f(x)$ has a **vertical asymptote** at $x = a$ if either* $\lim_{x\to a-} f(x) = \pm\infty$ *or* $\lim_{x\to a+} f(x) = \pm\infty$.

Definition 4.5.2. *The graph of $y = f(x)$ has a **horizontal asymptote** at $y = L$ if either* $\lim_{x\to\infty} f(x) = L$ *or* $\lim_{x\to-\infty} f(x) = L$.

Example 4.5.1. *Find the vertical and the horizontal asymptotes of* $f(x) = \frac{1}{x^2 - x}$.

Example 4.5.2. $f(x) = e^x$ *has a left horizontal asymptote* $y = 0$, $\lim_{x\to-\infty} e^x = 0$.

Example 4.5.3. $f(x) = \tan^{-1} x$ *has a two one sided limits,* $\lim_{x\to\infty} \tan^{-1} x = \pi/2$ *and* $\lim_{x\to-\infty} \tan^{-1} x = -\pi/2$.

Example 4.5.4. *Let* $f(x) = \frac{x^2+1}{x} = x + \frac{1}{x}$. *Then* $\lim_{x\to\pm\infty}(f(x) - x) = 0$. *Hence* f *has a two-sided oblique asymptote.*

Asymptotes of rational function

Let $f(x) = \frac{P_m(x)}{Q_n(x)}$, where P_m and Q_n are polynomials of degree m and n respectively. Suppose that P_m and Q_n have no common linear factors. The graph of f has

1. a vertical asymptote at every position at every x for which $Q_n(x) = 0$.
2. a two-sided horizontal asymptote $y = 0$ only if $m < n$.
3. a two-sided horizontal asymptote $y = L$ only if $m = n$. L is the ratio of the coefficients of the highest degree terms in P_m and Q_n.
4. a two sided oblique asymptote only if $m = n + 1$.

Example 4.5.5. *Find the oblique asymptote of* $y = \frac{x^3}{x^2+x+1}$.

Checklist For Curve Sketching

1. Examine $f(x)$ to find the domain, intercepts, asymptotes and even/odd symmetries.
2. Find points where $f'(x) = 0$ (critical points of f) and where $f'(x)$ is undefined (singular points of f).
3. Find points where $f''(x) = 0$ (critical points of f) and where $f''(x)$ is undefined (singular points of f).
4. Make a table to investigate the signs of $f'(x)$ and $f''(x)$ to find the intervals where f is increasing or decreasing and the intervals where f is concave up and down. Find also the extreme points and inflection points of the graph.

4.6 Extreme Value Problems

Example 4.6.1. *Find the area of the largest rectangle that can be inscribed in a semicircle of radius R if one side of the rectangle lies along the diameter of the semicircle.*

Example 4.6.2. *Find the shortest distance from the origin to the curve* $x^2y^4 = 1$.

Example 4.6.3. *A manufacturer has 100 tons of metal that he can sell now with a profit of \$5 a ton. For each week that he delays shipment, he can produce another 10 tons of metal. However, for each week he waits, the profit drops 25 cents a ton. If he can sell the metal at any time, when is the best time to sell so that his profit is maximized?*

Exercises.

1. *Find the shortest distance from the point* $(8,1)$ *to the curve* $y = 1 + x^{3/2}$.

2. *Find the largest possible perimeter of the rectangle that can be inscribed in a semicircle of radius R if one side of the rectangle lies along the diameter of the semicircle.*

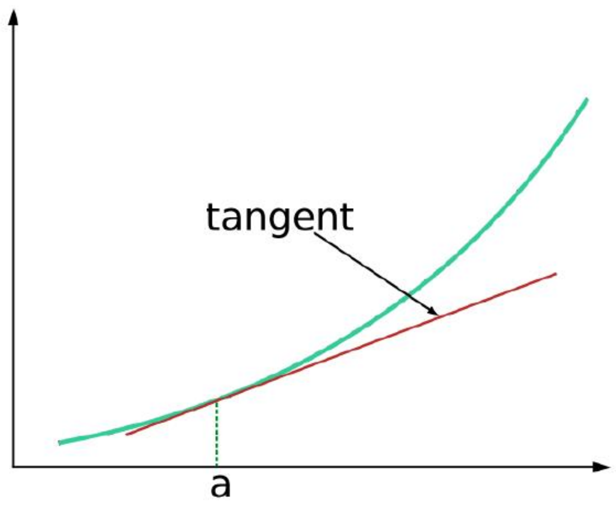
3. *Among all rectangles of perimeter P, show that the square has the greatest area.*

4. *Among all rectangles of given area A, show that the square has the least perimeter.*

5. *Find the equation of the straight line of maximum slope tangent to the curve* $y = 1 + 2x - x^3$.

4.7 Linear Approximation

The best line approximating the graph of $y = f(x)$ near $(a, f(a))$ is the tangent line through $(a, f(a))$.



The linearization of the function f about a is the function L defined by

$$L(x) = f(a) + f'(a)(x-a)$$

We say that L approximates f near $x = a$ and write $f(x) \approx L(x)$.

Example 4.7.1. *Using the linearization, approximate* $\sqrt{26}$*. (Hint: use the linearization of* $\sqrt{x}$ *at* $x = 25$*.)*

Example 4.7.2. *Approximate* $\cos \pi/5 = \cos 36°$ *using the linearization of* $\cos x$ *at* $x = \pi/6$.

Error Estimation

The error in the linear approximation is

$$\frac{f''(s)}{2}(x-a)^2$$

where s is some number between a and x. (The proof depends on the generalized mean value theorem.)

Since we do not know s, we have to choose $f''(s)$ to be largest (in absolute value) possible value, to get the maximum error.

So for the previous example, $f''(x) = -\sin x$, $a = \pi/6$, $x = \pi/5$, $\pi/6 < s < \pi/5$. Note that $f''(s) \leq 1$. So the error is smaller that $\frac{1}{2}(x-a)^2 = \frac{\pi^2}{1800} < 0.00549$. So $0.81367 - 0.00549 < \cos\pi/5 < 0.81367 + 0.00549$.

Exercises.

1. *Sketch* $y = \frac{1}{\sqrt{x}}$ *and its linearization about* $x = 4$.

2. *Approximate* $\sqrt{50}$ *using linearization.*

3. *Approximate* $\sin 46^\circ$ *using linearization.*

Sample Exam 2

1. Find the dimensions of the right triangle with hypotenuse $h = 2$ and maximum area.

2. Find the min and max values of $f(x) = 2x^3 - 15x^2 + 24x + 19$ on the interval $[0,5]$

3. Let $f(x) = x^4 - 3x^2 + 2$.

a) Find the intervals on which f is increasing or decreasing. Find all local extrema for f.

b) Find the intervals on which f is concave up or down. Find all inflection points for f.

c) Sketch a graph of $f(x)$ using parts (a) and (b).

4. Find all the horizontal and vertical asymptotes for

$$y = \frac{x^2 + 3}{1 - 3x^2}.$$

5. Let $f(x) = 2x^2 + x^3, x > 0$. Show that f is invertible and find $(f^{-1})'(16)$.

6. Find dy/dx by implicit differentiation if

$$y^2 e^x + y \ln x = 2.$$

7. Let $y = (1/x)^{\ln x}$. Find dy/dx.

8. Find $\cos(\sin^{-1} 0.7)$.